

AWSRaid

AWS CloudTrail Incident Investigation Report

Authentication compromise • S3 data access • IAM persistence

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Platform	CyberDefenders
Lab	AWSRaid
Category	Cloud Forensics / Incident Response
Primary tool	Splunk Enterprise
Evidence	AWS CloudTrail events
Report date	25 September 2026
Assessment	High severity

Purpose

Document the evidence, analytical method, findings, and defensive lessons from the supplied AWS CloudTrail dataset. All conclusions are bounded by the available lab evidence.

1. Executive Summary

The investigation identified a rapid cloud-account compromise followed by S3 reconnaissance, object retrieval, a security-relevant S3 configuration change, and IAM persistence. The strongest authentication anomaly involved the IAM user `helpdesk.luke` and source IP `185.192.70.84`: nine failed AWS console sign-ins occurred in 33 seconds, followed by a successful login four seconds after the last failure.

After authentication, the compromised identity enumerated S3 buckets, retrieved eight objects from seven buckets, disabled four bucket-level S3 Block Public Access safeguards on `backup-and-restore98825501`, created the IAM user `marketing.mark`, and added that user to the Admins group five seconds later.

Incident assessment — High

Confirmed unauthorized access to cloud-stored objects, weakening of storage safeguards, and creation of a privileged persistence account. The evidence includes access to backup, contract, customer-data, design, and secret-related objects.

2. Key Findings

Finding	Result
Compromised IAM user	<code>helpdesk.luke</code>
Primary suspicious login IP	<code>185.192.70.84</code>
First S3 object access	2023-11-02 09:55:53 UTC (09:55 in lab format)
Bucket containing the DWG file	<code>product-designs-repository31183937</code>
Bucket with modified public-access safeguards	<code>backup-and-restore98825501</code>
Attacker-created IAM user	<code>marketing.mark</code>
Group assigned to the new user	Admins

3. Scope and Objectives

The evidence source was a Splunk index named `aws_cloudtrail` containing 4,032 AWS CloudTrail events. The objectives were to identify the compromised identity, reconstruct its activity, determine the affected S3 resources, identify persistence actions, and translate the findings into detection and remediation guidance.

- Identify the compromised IAM user and suspicious authentication source.
- Determine the first confirmed access to an S3 object.
- Identify the S3 bucket containing a DWG design file.
- Identify the bucket whose public-access protections were modified.
- Recover the IAM account created for persistence and its assigned group.
- Build a defensible timeline and map the activity to MITRE ATT&CK.

4. Investigation Methodology

The investigation used a broad-to-narrow approach. Each step answered one analytical question and provided the pivot for the next step.

Step	Analytical question	CloudTrail/Splunk evidence
1	What services dominate the dataset?	Count events by eventSource.
2	Which authentication pattern is abnormal?	Review signin.amazonaws.com and aggregate failures.
3	Did a success follow the failures?	Sort ConsoleLogin events chronologically.
4	What did the identity do after login?	Pivot on username and suspicious source IP.
5	Was S3 only enumerated or were objects accessed?	Distinguish ListBuckets from GetObject.
6	Were storage controls changed?	Review PutBucketPublicAccessBlock and related calls.
7	Was persistence created?	Review CreateUser and AddUserToGroup.

4.1 Evidence interpretation rules

- Identity attribution used `userIdentity.userName` and `userIdentity.arn`; IP alone was not treated as identity proof.
- A blank `errorCode/errorMessage` field, combined with returned response fields, was treated as evidence that the API call succeeded.
- `eventName` determined the action: discovery calls were not counted as data access.
- `requestParameters` identified the target bucket, object key, username, or group.
- `eventTime` values ending in Z were interpreted as UTC.
- Inferences are explicitly labeled where the evidence does not prove intent or infrastructure ownership.

5. Initial CloudTrail Triage

SPL — initial CloudTrail dataset

```
index="aws_cloudtrail"
```

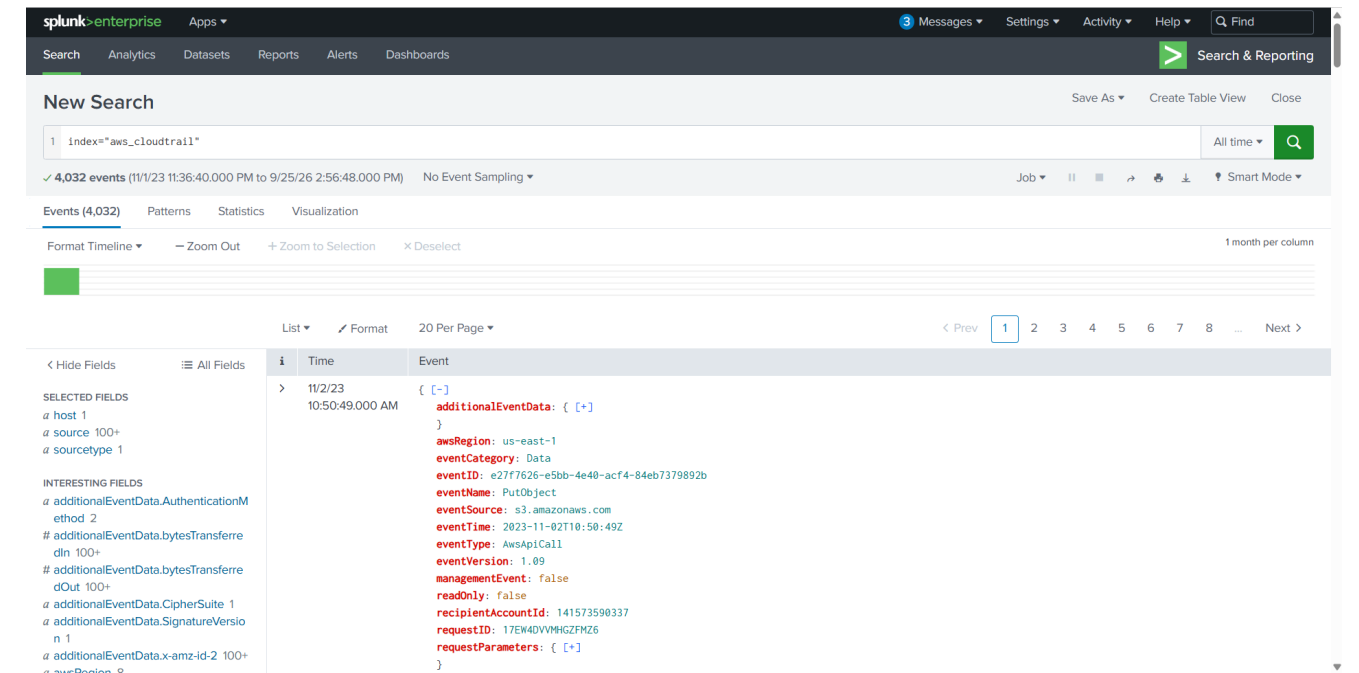


Figure 1. Initial review of the 4,032 CloudTrail events in Splunk.

SPL — event volume by AWS service

```
index="aws_cloudtrail"
| stats count by eventSource
| sort - count
```

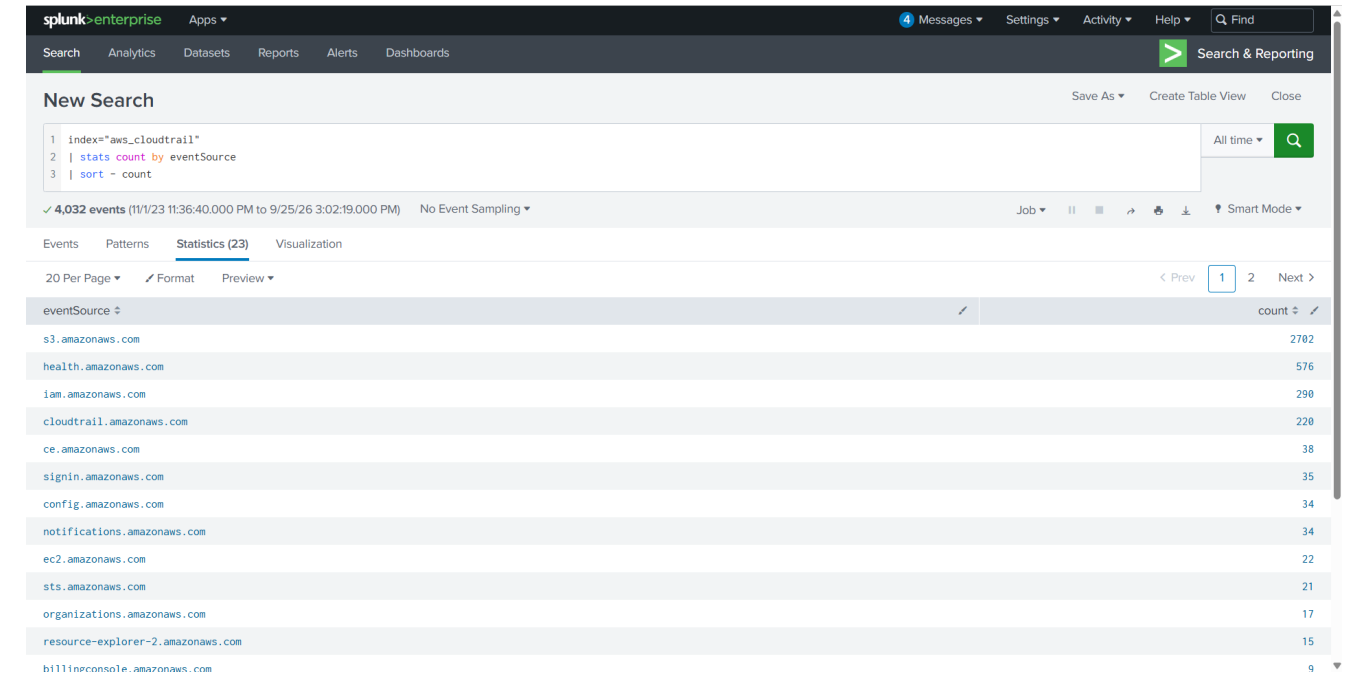


Figure 2. CloudTrail events grouped by AWS event source.

The query produced 23 event sources. S3 generated 2,702 events, followed by AWS Health (576), IAM (290), CloudTrail (220), and AWS Sign-In (35). Event volume is a triage signal, not proof of compromise: S3 was prioritized because it dominated the dataset, while sign-in and IAM were prioritized because they represent identity and persistence activity.

6. Finding 1 — Compromised IAM User

6.1 Why authentication was examined first

Console authentication is a logical starting point because a compromised cloud identity can access many services without exploiting a host. The `signin.amazonaws.com` event source isolates AWS Management Console authentication activity.

SPL — chronological console logins

```
index="aws_cloudtrail" eventSource="signin.amazonaws.com"
| table _time eventName userIdentity.arn userIdentity.userName sourceIPAddress userAgent
"responseElements.ConsoleLogin" errorMessage
| sort 0 _time
```

_time	eventName	userIdentity.arn	userIdentity.userName	sourceIPAddress	userAgent	responseElements.ConsoleLogin	errorMessage
2023-11-02 09:44:08	ConsoleLogin	arn:aws:iam::141573590337:user/itmanager.david	itmanager.david	194.5.53.98	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/118.0.0.0 Safari/537.36	Success	
2023-11-02 09:53:27	ConsoleLogin		helpdesk.luke	185.192.70.84	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/118.0.0.0 Safari/537.36	Failure	Failed authentication
2023-11-02 09:53:34	ConsoleLogin		helpdesk.luke	185.192.70.84	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/118.0.0.0 Safari/537.36	Failure	Failed authentication
2023-11-02 09:53:36	ConsoleLogin		helpdesk.luke	185.192.70.84	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/118.0.0.0 Safari/537.36	Failure	Failed authentication
2023-11-02 09:53:39	ConsoleLogin		helpdesk.luke	185.192.70.84	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/118.0.0.0 Safari/537.36	Failure	Failed authentication
2023-11-02 09:53:42	ConsoleLogin		helpdesk.luke	185.192.70.84	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/118.0.0.0 Safari/537.36	Failure	Failed authentication
2023-11-02 09:53:46	ConsoleLogin		helpdesk.luke	185.192.70.84	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/118.0.0.0 Safari/537.36	Failure	Failed authentication
2023-11-02 09:53:51	ConsoleLogin		helpdesk.luke	185.192.70.84	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/118.0.0.0 Safari/537.36	Failure	Failed authentication
2023-11-02 09:53:55	ConsoleLogin		helpdesk.luke	185.192.70.84	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/118.0.0.0 Safari/537.36	Failure	Failed authentication
2023-11-02 09:54:00	ConsoleLogin		helpdesk.luke	185.192.70.84	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/118.0.0.0 Safari/537.36	Failure	Failed authentication
2023-11-02 09:54:04	ConsoleLogin	arn:aws:iam::141573590337:user/helpdesk.luke	helpdesk.luke	185.192.70.84	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/118.0.0.0 Safari/537.36	Success	
2023-11-02 10:11:24	ConsoleLogin		sysadmin.mary	173.239.214.84	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/118.0.0.0 Safari/537.36	Failure	Failed authentication
2023-11-02 10:11:35	ConsoleLogin	arn:aws:iam::141573590337:user/sysadmin.mary	sysadmin.mary	173.239.214.84	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/118.0.0.0 Safari/537.36	Success	
2023-11-02 10:12:11	ConsoleLogin	arn:aws:iam::141573590337:user/marketing.sophia	marketing.sophia	173.239.214.120	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/118.0.0.0 Safari/537.36	Success	

Figure 3. Chronological authentication sequence showing failures followed by success.

The table command exposes the fields required to answer who authenticated, from where, with which client, and whether the attempt succeeded. The `sort 0 _time` command preserves the entire result set and orders it from earliest to latest.

6.2 Aggregating failed attempts

SPL — failed logins by identity and source

```
index="aws_cloudtrail"
eventSource="signin.amazonaws.com"
"responseElements.ConsoleLogin"="Failure"
| stats count AS failed_attempts
   min(_time) AS first_seen
   max(_time) AS last_seen
   values(userAgent) AS user_agents
   by userIdentity.userName sourceIPAddress
| convert ctime(first_seen) ctime(last_seen)
| sort - failed_attempts
```

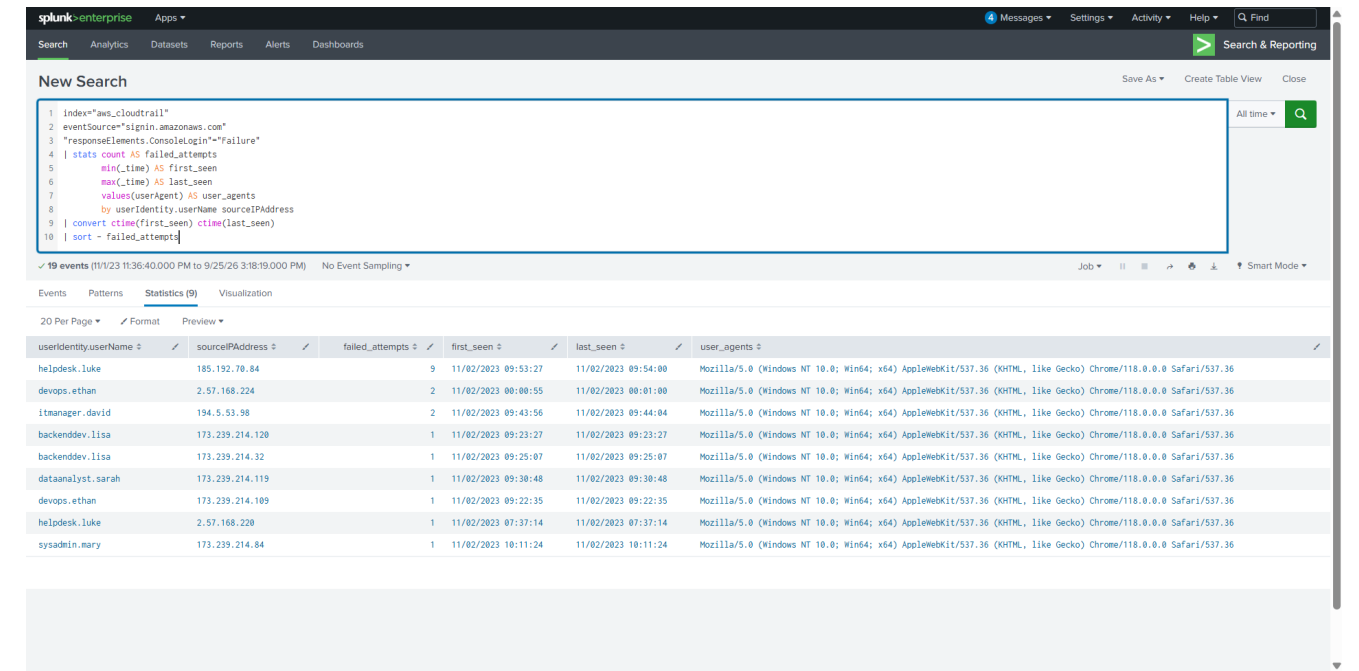


Figure 4. Failed console logins aggregated by user and source IP.

The largest cluster consisted of nine failed attempts against helpdesk.luke from 185.192.70.84 between 09:53:27 and 09:54:00 UTC. The next event for the same identity and source was a successful ConsoleLogin at 09:54:04 UTC.

Metric	Value
Failed attempts	9
Failure window	33 seconds
Last failure	2023-11-02 09:54:00 UTC
Successful login	2023-11-02 09:54:04 UTC
Gap to success	4 seconds
User agent	Windows 10 / Chrome 118

Analytical conclusion

The compromised account was helpdesk.luke. The sequence is strongly consistent with password guessing or use of newly obtained credentials. CloudTrail confirms the failures and success but does not reveal exactly how the correct password was acquired.

7. Scope Pivot — Activity After Authentication

SPL — compromised identity and suspicious-IP scope

```
index="aws_cloudtrail"
(sourceIPAddress="185.192.70.84" OR userIdentity.userName="helpdesk.luke")
| table _time eventSource eventName userIdentity.userName userIdentity.arn sourceIPAddress userAgent errorCode
errorMessage
| sort 0 _time
```

_time	eventSource	eventName	userIdentity.userName	userIdentity.arn	sourceIPAddress	userAgent	errorCode	errorMessage
2023-11-02 09:54:00	signin.amazonaws.com	ConsoleLogin	helpdesk.luke		185.192.70.84	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/118.0.0.0 Safari/537.36		Failed authentication
2023-11-02 09:54:04	signin.amazonaws.com	ConsoleLogin	helpdesk.luke	arn:aws:iam::141573590337:user/helpdesk.luke	185.192.70.84	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/118.0.0.0 Safari/537.36		
2023-11-02 09:54:38	ce.amazonaws.com	GetCostForecast	helpdesk.luke	arn:aws:iam::141573590337:user/helpdesk.luke	185.192.70.90	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/118.0.0.0 Safari/537.36	AccessDenied	IAM user access not activated
2023-11-02 09:54:38	ce.amazonaws.com	GetCostAndUsage	helpdesk.luke	arn:aws:iam::141573590337:user/helpdesk.luke	185.192.70.90	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/118.0.0.0 Safari/537.36	AccessDenied	IAM user access not activated
2023-11-02 09:54:39	health.amazonaws.com	DescribeEventAggregates	helpdesk.luke	arn:aws:iam::141573590337:user/helpdesk.luke	185.192.70.79	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/118.0.0.0 Safari/537.36		
2023-11-02 09:54:39	health.amazonaws.com	DescribeEventAggregates	helpdesk.luke	arn:aws:iam::141573590337:user/helpdesk.luke	health.amazonaws.com	health.amazonaws.com		
2023-11-02 09:54:39	health.amazonaws.com	DescribeEventAggregates	helpdesk.luke	arn:aws:iam::141573590337:user/helpdesk.luke	185.192.70.74	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/118.0.0.0 Safari/537.36		
2023-11-02 09:54:39	health.amazonaws.com	DescribeEventAggregates	helpdesk.luke	arn:aws:iam::141573590337:user/helpdesk.luke	185.192.70.3	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/118.0.0.0 Safari/537.36		
2023-11-02 09:54:39	health.amazonaws.com	DescribeEventAggregates	helpdesk.luke	arn:aws:iam::141573590337:user/helpdesk.luke	health.amazonaws.com	health.amazonaws.com		

Figure 5. The 480 events associated with the suspicious IP or compromised identity.

The pivot returned 480 events. Including the username was essential because subsequent activity did not remain on the original login IP. The same IAM identity appeared from several addresses in the 185.192.70.x range during the incident window.

IP-rotation inference

The close timing, common IAM ARN, and consistent client characteristics suggest rotating proxy/VPN infrastructure or lab-generated address rotation. This is an inference; the available data does not prove ownership of the IPs. Values such as health.amazonaws.com in sourceIPAddress represent AWS service-originated activity and should not be treated as attacker endpoints.

8. Finding 2 — First S3 Object Access

8.1 Discovery is not object access

The compromised identity first called ListBuckets at 09:55:09 UTC. This enumerated available S3 buckets and established discovery behavior, but it did not retrieve a stored object.

SPL — S3 enumeration

```
index="aws_cloudtrail"
eventSource="s3.amazonaws.com"
eventName="ListBuckets"
userIdentity.userName="helpdesk.luke"
| table _time eventTime eventName userIdentity.arn sourceIPAddress userAgent errorCode errorMessage
| sort 0 _time
```

New Search

```

1 index="aws:cloudtrail"
2 (sourceIPAddress="185.192.70.84" OR userIdentity.userName="helpdesk.luke")
3 | table _time eventSource eventName userIdentity.arn sourceIPAddress userAgent errorCode errorMessage
4 | sort _time

```

✓ 480 events (11/23 11:36:40.000 PM to 9/25/26 3:46:36.000 PM) No Event Sampling

Events Patterns **Statistics (480)** Visualization

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_time	eventSource	eventName	userIdentity.userName	userIdentity.arn	sourceIPAddress	userAgent	errorCode	errorMessage
2023-11-02 09:54:40	health.amazonaws.com	DescribeEventAggregates	helpdesk.luke	arn:aws:iam::141573590337:user/helpdesk.luke	185.192.70.74	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/118.0.0.0 Safari/537.36		
2023-11-02 09:54:40	health.amazonaws.com	DescribeEventAggregates	helpdesk.luke	arn:aws:iam::141573590337:user/helpdesk.luke	health.amazonaws.com	health.amazonaws.com		
2023-11-02 09:54:40	health.amazonaws.com	DescribeEventAggregates	helpdesk.luke	arn:aws:iam::141573590337:user/helpdesk.luke	185.192.70.69	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/118.0.0.0 Safari/537.36		
2023-11-02 09:54:40	health.amazonaws.com	DescribeEventAggregates	helpdesk.luke	arn:aws:iam::141573590337:user/helpdesk.luke	185.192.70.72	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/118.0.0.0 Safari/537.36		
2023-11-02 09:54:40	health.amazonaws.com	DescribeEventAggregates	helpdesk.luke	arn:aws:iam::141573590337:user/helpdesk.luke	185.192.70.81	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/118.0.0.0 Safari/537.36		
2023-11-02 09:54:40	health.amazonaws.com	DescribeEventAggregates	helpdesk.luke	arn:aws:iam::141573590337:user/helpdesk.luke	health.amazonaws.com	health.amazonaws.com		
2023-11-02 09:54:40	health.amazonaws.com	DescribeEventAggregates	helpdesk.luke	arn:aws:iam::141573590337:user/helpdesk.luke	health.amazonaws.com	health.amazonaws.com		
2023-11-02 09:54:44	ec2.amazonaws.com	DescribeRegions	helpdesk.luke	arn:aws:iam::141573590337:user/helpdesk.luke	185.192.70.99	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/118.0.0.0 Safari/537.36		
2023-11-02 09:54:47	resource-explorer-2.amazonaws.com	ListIndexes	helpdesk.luke	arn:aws:iam::141573590337:user/helpdesk.luke	185.192.70.72	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/118.0.0.0 Safari/537.36		
2023-11-02 09:54:48	resource-explorer-2.amazonaws.com	ListIndexes	helpdesk.luke	arn:aws:iam::141573590337:user/helpdesk.luke	185.192.70.72	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/118.0.0.0 Safari/537.36		
2023-11-02 09:55:09	s3.amazonaws.com	ListBuckets	helpdesk.luke	arn:aws:iam::141573590337:user/helpdesk.luke	185.192.70.78	[S3Console/0.4, aws-internal/3 aws-sdk-java/1.12.488 Linux/5.10.197-164.748.amzn2int.x86_64 OpenJDK-64-Bit_Server-30/26.372-0408 java/1.8.0_372 vendor/OracleCorporation cfg/retry-mode/standard]		

Figure 6. First S3 enumeration visible in the scoped activity timeline.

8.2 Confirming object retrieval

SPL — S3 object access

```

index="aws:cloudtrail"
eventSource="s3.amazonaws.com"
eventName="GetObject"
userIdentity.userName="helpdesk.luke"
| table _time eventTime eventName userIdentity.arn sourceIPAddress requestParameters.bucketName requestParameters.key errorCode errorMessage
| sort 0 _time

```

New Search

```

1 index="aws:cloudtrail"
2 eventSource="s3.amazonaws.com"
3 eventName="GetObject"
4 userIdentity.userName="helpdesk.luke"
5 | table _time eventTime eventName userIdentity.arn sourceIPAddress requestParameters.bucketName requestParameters.key errorCode errorMessage
6 | sort 0 _time

```

✓ 8 events (11/23 11:36:40.000 PM to 9/25/26 3:49:06.000 PM) No Event Sampling

Events Patterns **Statistics (8)** Visualization

20 Per Page Format Preview

_time	eventTime	eventName	userIdentity.arn	sourceIPAddress	requestParameters.bucketName	requestParameters.key	errorCode	errorMessage
2023-11-02 09:55:53	2023-11-02T09:55:53Z	GetObject	arn:aws:iam::141573590337:user/helpdesk.luke	185.192.70.71	research-project-files23411723	prototype.obj		
2023-11-02 09:56:07	2023-11-02T09:56:07Z	GetObject	arn:aws:iam::141573590337:user/helpdesk.luke	185.192.70.3	product-designs-repository31183937	Product2_CAD_Designs.dwg		
2023-11-02 09:56:20	2023-11-02T09:56:20Z	GetObject	arn:aws:iam::141573590337:user/helpdesk.luke	185.192.70.83	marketing-assets-vault27512203	logo.png		
2023-11-02 09:56:30	2023-11-02T09:56:30Z	GetObject	arn:aws:iam::141573590337:user/helpdesk.luke	185.192.70.86	legal-docs45920393	Contract_Agreement.pdf		
2023-11-02 09:56:39	2023-11-02T09:56:39Z	GetObject	arn:aws:iam::141573590337:user/helpdesk.luke	185.192.70.93	customer-data-backup57893984	CustomerData_Backup_2023-11-01.zip		
2023-11-02 09:56:50	2023-11-02T09:56:50Z	GetObject	arn:aws:iam::141573590337:user/helpdesk.luke	185.192.70.101	contracts-data67988444	Contract_Termination_Letter_Client.pdf		
2023-11-02 09:57:04	2023-11-02T09:57:04Z	GetObject	arn:aws:iam::141573590337:user/helpdesk.luke	185.192.70.89	backup-and-restore98825501	Configuration_Backup_Server2.zip		
2023-11-02 09:57:11	2023-11-02T09:57:11Z	GetObject	arn:aws:iam::141573590337:user/helpdesk.luke	185.192.70.90	backup-and-restore98825501	secrets_vault_dump.bak		

Figure 7. Eight GetObject events attributed to helpdesk.luke.

The earliest GetObject event occurred at 2023-11-02 09:55:53 UTC and retrieved prototype.obj from research-project-files23411723. Because the lab requested YYYY-MM-DD HH:MM, the submitted answer is 2023-11-02 09:55.

Key distinction

09:55:09 marks S3 enumeration (ListBuckets). 09:55:53 marks the first confirmed S3 object access (GetObject). Selecting the earlier timestamp would answer a different question.

9. Finding 3 — Accessed Objects and DWG Bucket

The compromised identity retrieved eight objects from seven buckets in approximately 78 seconds. The rapid cross-bucket sequence suggests targeted collection rather than ordinary interactive browsing.

UTC	Bucket	Object key
09:55:53	research-project-files23411723	prototype.obj
09:56:07	product-designs-repository31183937	Product2_CAD_Designs.dwg
09:56:20	marketing-assets-vault27512203	logo.png
09:56:30	legal-docs45020393	Contract_Agreement.pdf
09:56:39	customer-data-backup57893984	CustomerData_Backup_2023-11-01.zip
09:56:50	contracts-data67988444	Contract_Termination_Letter_Client.pdf
09:57:04	backup-and-restore98825501	Configuration_Backup_Server2.zip
09:57:11	backup-and-restore98825501	secrets_vault_dump.bak

DWG is a native AutoCAD drawing format commonly used for engineering, architecture, and product-design data. Filtering object keys by extension isolated the relevant access.

SPL — DWG object access

```
index="aws_cloudtrail"  
eventSource="s3.amazonaws.com"  
eventName="GetObject"  
userIdentity.userName="helpdesk.luke"  
requestParameters.key="*.dwg"  
| table _time requestParameters.bucketName requestParameters.key sourceIPAddress
```

New Search

```

1 index="aws:cloudtrail"
2 eventSource="s3.amazonaws.com"
3 eventName="GetObject"
4 userIdentity.userName="helpdesk.luke"
5 | table _time eventTime eventName userIdentity.arn sourceIPAddress requestParameters.bucketName requestParameters.key errorCode errorMessage
6 | sort 0 _time

```

8 events (11/23 11:36:40 PM to 9/25/26 3:49:06 PM) No Event Sampling

Events Patterns **Statistics (8)** Visualization

20 Per Page Format Preview

_time	eventTime	eventName	userIdentity.arn	sourceIPAddress	requestParameters.bucketName	requestParameters.key	errorCode	errorMessage
2023-11-02 09:55:53	2023-11-02T09:55:53Z	GetObject	arn:aws:iam::141573590337:user/helpdesk.luke	185.192.70.71	research-project-files23411723	prototype.obj		
2023-11-02 09:56:07	2023-11-02T09:56:07Z	GetObject	arn:aws:iam::141573590337:user/helpdesk.luke	185.192.70.3	product-designs-repository31183937	Product2_CAD_Designs.dwg		
2023-11-02 09:56:20	2023-11-02T09:56:20Z	GetObject	arn:aws:iam::141573590337:user/helpdesk.luke	185.192.70.83	marketing-assets-vault27512203	logo.png		
2023-11-02 09:56:30	2023-11-02T09:56:30Z	GetObject	arn:aws:iam::141573590337:user/helpdesk.luke	185.192.70.86	legal-docs45020393	Contract_Agreement.pdf		
2023-11-02 09:56:39	2023-11-02T09:56:39Z	GetObject	arn:aws:iam::141573590337:user/helpdesk.luke	185.192.70.93	customer-data-backup57893984	CustomerData_Backup_2023-11-01.zip		
2023-11-02 09:56:50	2023-11-02T09:56:50Z	GetObject	arn:aws:iam::141573590337:user/helpdesk.luke	185.192.70.101	contracts-data67988444	Contract_Termination_Letter_Client.pdf		
2023-11-02 09:57:04	2023-11-02T09:57:04Z	GetObject	arn:aws:iam::141573590337:user/helpdesk.luke	185.192.70.89	backup-and-restore98825501	Configuration_Backup_Server2.zip		
2023-11-02 09:57:11	2023-11-02T09:57:11Z	GetObject	arn:aws:iam::141573590337:user/helpdesk.luke	185.192.70.90	backup-and-restore98825501	secrets_vault_dump.bak		

Figure 8. GetObject evidence containing Product2_CAD_Designs.dwg and its bucket.

Finding

Product2_CAD_Designs.dwg was retrieved from product-designs-repository31183937 at 09:56:07 UTC.

10. Finding 4 — S3 Public-Access Safeguards Modified

SPL — security-relevant S3 configuration changes

```

index="aws:cloudtrail"
eventSource="s3.amazonaws.com"
userIdentity.userName="helpdesk.luke"
(eventName="PutBucketPublicAccessBlock" OR eventName="DeletePublicAccessBlock" OR eventName="PutBucketAcl" OR
eventName="PutBucketPolicy")
| table _time eventName userIdentity.arn sourceIPAddress requestParameters.bucketName requestParameters
errorCode errorMessage
| sort 0 _time

```

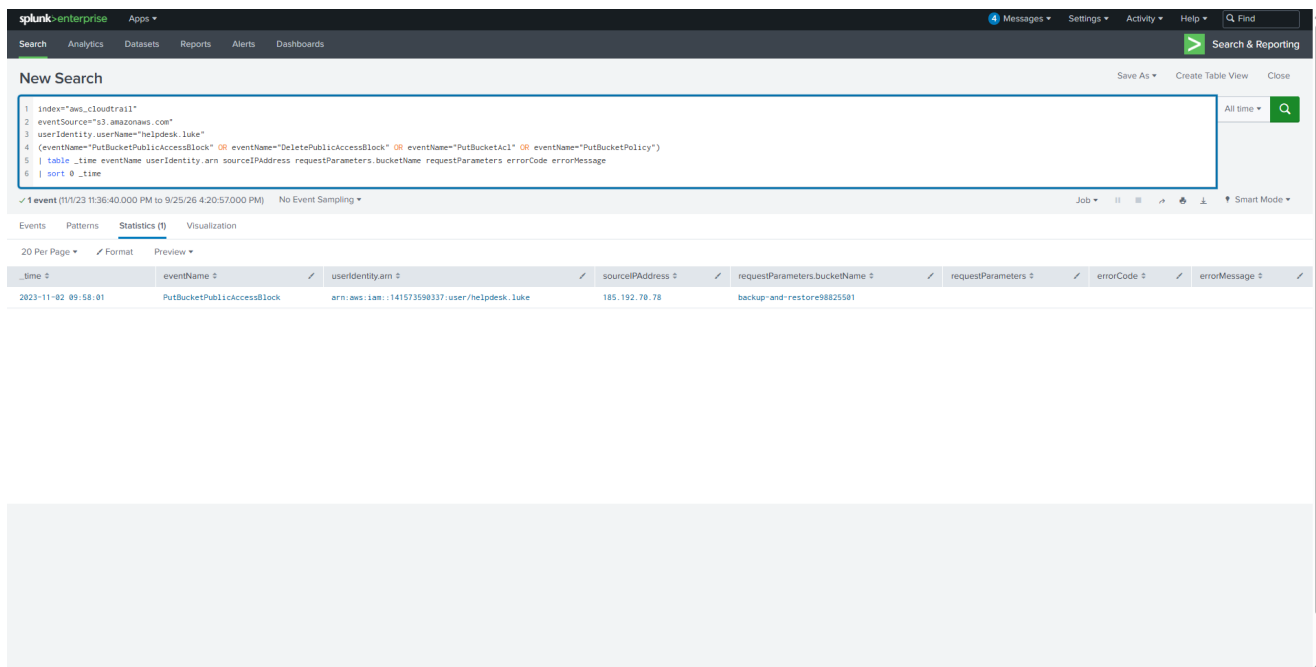


Figure 9. Query result identifying the affected bucket and PutBucketPublicAccessBlock event.

The matching event was expanded to verify the exact request parameters rather than relying only on the API name.

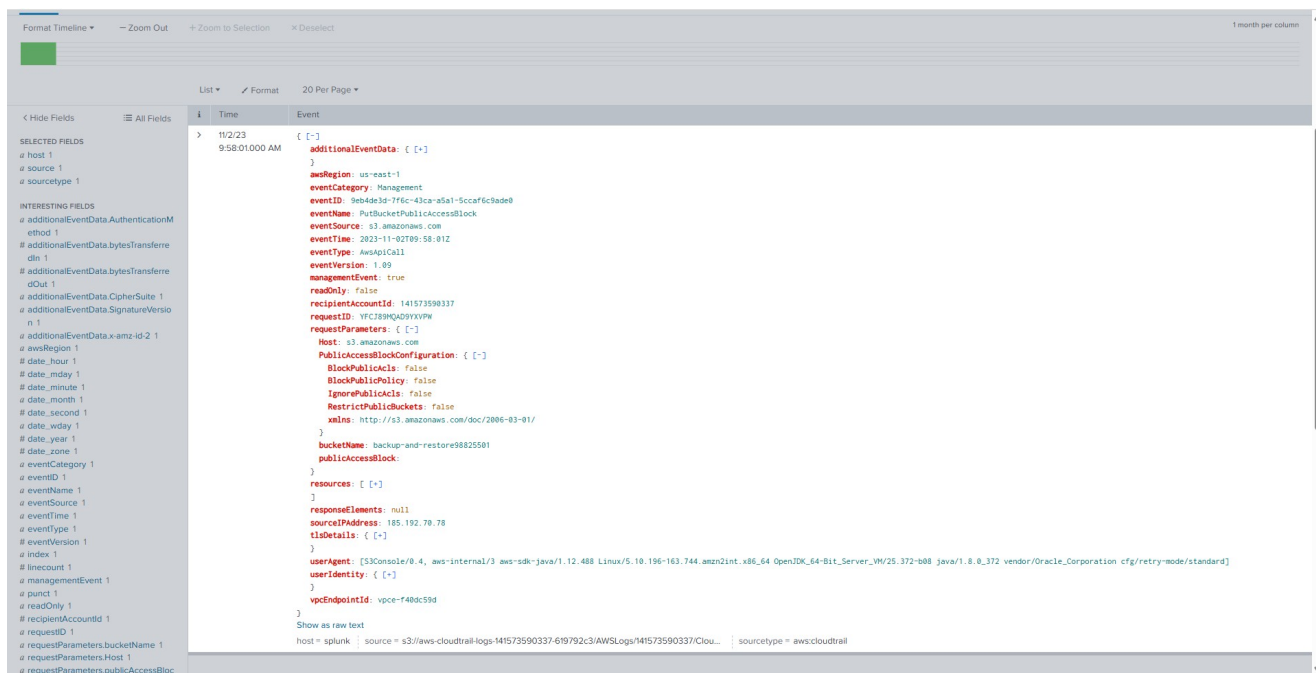


Figure 10. Expanded event showing all four S3 Block Public Access settings set to false.

At 09:58:01 UTC, helpdesk.luke issued PutBucketPublicAccessBlock against backup-and-restore98825501 from 185.192.70.78. The raw event recorded managementEvent=true, readOnly=false, no error, and the following request parameters:

Setting	Recorded value	Security effect
BlockPublicAcls	false	Public ACL submissions are no longer blocked by this bucket-level control.
IgnorePublicAcls	false	Existing or new public ACLs are no longer

Setting	Recorded value	Security effect
		ignored by this control.
BlockPublicPolicy	false	Public bucket policies are no longer rejected by this control.
RestrictPublicBuckets	false	Public-policy access is no longer restricted by this control.

Precision note

Setting all four bucket-level flags to false weakens the safeguards and permits a public ACL or policy to take effect. This event alone does not prove anonymous exposure because account-level controls and the effective ACL/policy were not provided.

Finding

The affected bucket was backup-and-restore98825501.

11. Finding 5 — Attacker-Created IAM User

IAM CreateUser events were reviewed because new cloud accounts are a common persistence mechanism. The query constrained results to the compromised actor and exposed both the requested username and the returned user ARN.

SPL — IAM user creation

```
index="aws_cloudtrail"
eventSource="iam.amazonaws.com"
userIdentity.userName="helpdesk.luke"
eventName="CreateUser"
| table _time eventName userIdentity.arn sourceIPAddress requestParameters.userName responseElements.user.arn
errorCode errorMessage
| sort 0 _time
```

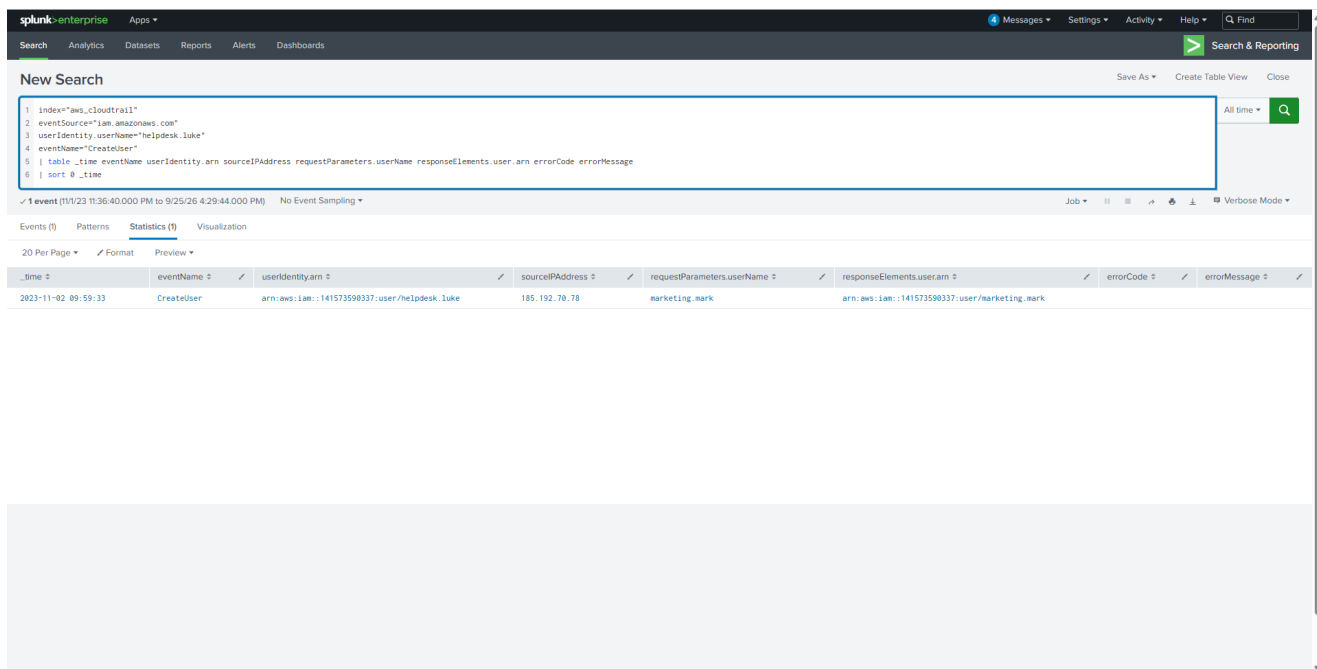


Figure 11. Successful creation of the IAM user marketing.mark.

At 09:59:33 UTC, helpdesk.luke created marketing.mark from 185.192.70.78. The returned ARN `arn:aws:iam::141573590337:user/marketing.mark` and empty error fields confirm success.

Finding

The attacker-created IAM user was marketing.mark.

12. Finding 6 — Privileged Group Assignment

Creating an account alone does not prove its privilege level. The next step was to search for AddUserToGroup and inspect requestParameters.groupName.

SPL — group membership change

```
index="aws_cloudtrail"
eventSource="iam.amazonaws.com"
eventName="AddUserToGroup"
requestParameters.userName="marketing.mark"
| table _time eventName userIdentity.arn sourceIPAddress requestParameters.userName
requestParameters.groupName errorCode errorMessage
| sort 0 _time
```

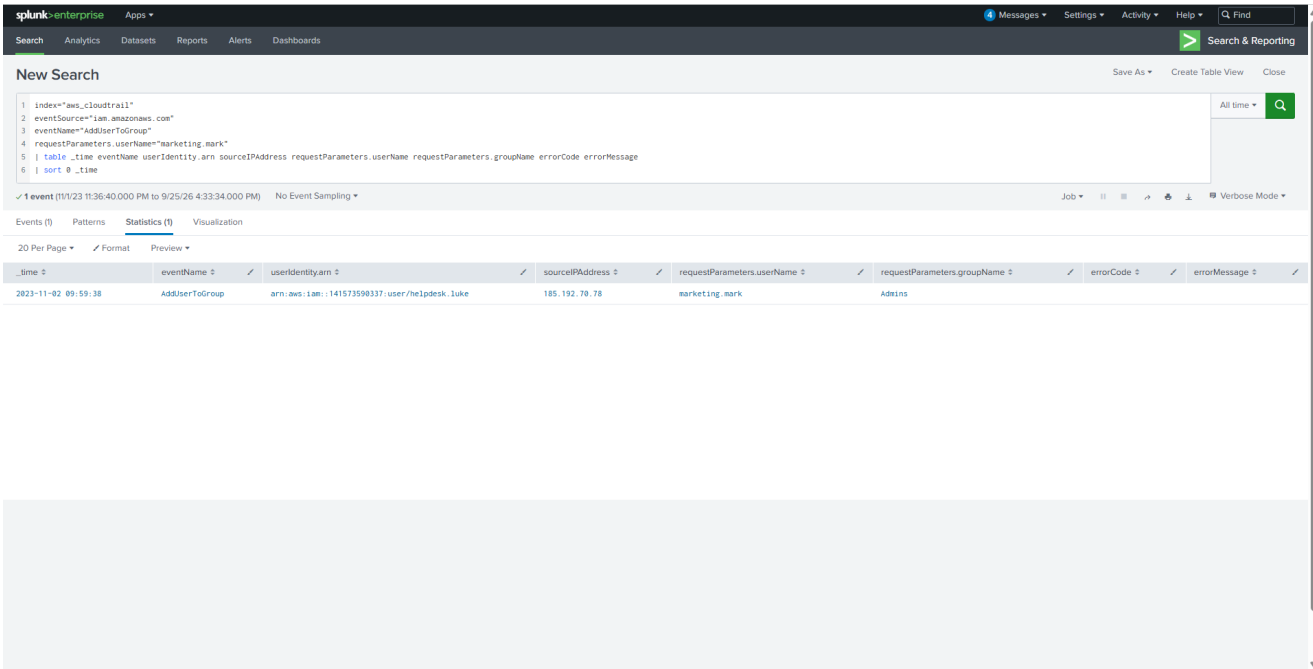


Figure 12. marketing.mark added to the Admins group.

At 09:59:38 UTC—five seconds after creation—helpdesk.luke added marketing.mark to Admins from the same source IP. The absence of an error indicates the action succeeded.

Finding

The new account was added to Admins, establishing a secondary privileged identity for persistence and likely privilege escalation.

13. Reconstructed Incident Timeline

Time (UTC)	CloudTrail event	Evidence-based interpretation
09:53:27–09:54:00	9 × ConsoleLogin Failure	Concentrated authentication failures against helpdesk.luke.
09:54:04	ConsoleLogin Success	Compromised identity successfully authenticated.
09:55:09	ListBuckets	S3 discovery began.
09:55:53	GetObject	First confirmed object access: prototype.obj.
09:56:07	GetObject	Product2_CAD_Designs.dwg retrieved.
09:57:11	GetObject	secrets_vault_dump.bak retrieved.
09:58:01	PutBucketPublicAccessBlock	Four bucket safeguards set to false.
09:59:33	CreateUser	marketing.mark created.
09:59:38	AddUserToGroup	marketing.mark added to Admins.

Speed of compromise

Only 5 minutes and 34 seconds elapsed between the successful console login and the creation of privileged persistence. The compact timeline supports high-confidence correlation across the authentication, S3, and IAM events.

14. Impact Assessment

Impact area	Evidence	Assessment
Confidentiality	Eight S3 GetObject operations	Confirmed access to design, legal, customer-backup, configuration, and secret-related files.
Integrity / security posture	S3 Block Public Access fields set false	Bucket-level safeguards were weakened; effective public exposure requires additional validation.
Persistence	CreateUser + AddUserToGroup	Secondary privileged IAM identity successfully established.
Credential risk	secrets_vault_dump.bak accessed	Potential downstream credential/key exposure; file contents were not available in the dataset.

15. Indicators of Compromise and Investigation Leads

Type	Value	Context
IAM user	helpdesk.luke	Compromised identity
IPv4	185.192.70.84	Primary suspicious console-login source
IPv4 pattern	185.192.70.x	Related activity observed during incident window

Type	Value	Context
IAM user	marketing.mark	Attacker-created persistence account
IAM group	Admins	Privileged group assigned to new user
S3 bucket	backup-and-restore98825501	Controls weakened; two sensitive objects accessed
Object key	Product2_CAD_Designs.dwg	Product-design data
Object key	secrets_vault_dump.bak	Potential secret material

16. MITRE ATT&CK Mapping

The mapping below connects observed behavior to ATT&CK. T1110.001 is presented as a suspected technique because CloudTrail shows the pattern but not the credential-acquisition method.

Tactic	Technique	Evidence
Credential Access	T1110.001 — Password Guessing	Nine failures followed by a success in seconds; technique is strongly indicated, not cryptographically proven.
Initial Access / Persistence	T1078.004 — Valid Accounts: Cloud Accounts	Successful console use of helpdesk.luke.
Discovery	T1619 — Cloud Storage Object Discovery	ListBuckets before targeted GetObject activity.
Collection	T1530 — Data from Cloud Storage	Eight successful S3 object retrievals.
Persistence	T1136.003 — Create Account: Cloud Account	marketing.mark created through IAM.
Persistence / Privilege Escalation	T1098.003 — Additional Cloud Roles	New user added to Admins.

17. Detection Engineering Opportunities

- Alert on repeated ConsoleLogin failures for one account followed by Success within five minutes.
- Correlate a new or unusual console source IP with immediate ListBuckets and GetObject activity.
- Detect rapid GetObject access across multiple buckets, particularly for backup, secret, contract, customer, and design file patterns.
- Alert whenever any PublicAccessBlockConfiguration field is set to false.
- Raise a critical alert for CreateUser followed by AddUserToGroup for a privileged group within a short window.
- Maintain per-user baselines for source network, user agent, AWS service, and time of day.

17.1 High-value Splunk detections

Detection — failures followed by success

```
index="aws_cloudtrail" eventSource="signin.amazonaws.com"
| eval result=coalesce('responseElements.ConsoleLogin', "Unknown")
| stats count(eval(result="Failure")) AS failures
           count(eval(result="Success")) AS successes
           min(_time) AS first_seen max(_time) AS last_seen
           by userIdentity.userName sourceIPAddress
| where failures >= 5 AND successes >= 1 AND (last_seen-first_seen) <= 300
```

Detection — S3 public-access safeguards disabled

```
index="aws_cloudtrail" eventSource="s3.amazonaws.com" eventName="PutBucketPublicAccessBlock"
| search requestParameters.publicAccessBlockConfiguration.*=false
| table _time userIdentity.arn sourceIPAddress requestParameters.bucketName
requestParameters.publicAccessBlockConfiguration.*
```

Detection — user creation followed by group assignment

```
index="aws_cloudtrail" eventSource="iam.amazonaws.com"
(eventName="CreateUser" OR eventName="AddUserToGroup")
| transaction requestParameters.userName maxspan=10m
| search eventcount>=2
| table _time userIdentity.arn sourceIPAddress requestParameters.userName requestParameters.groupName duration
```

18. Containment and Remediation

1. Disable helpdesk.luke, revoke active sessions, and rotate its console password, access keys, and any related credentials.
2. Remove marketing.mark and review every IAM user, group, policy, role, login profile, access key, and token created or changed during the incident window.
3. Restore all four S3 Block Public Access settings at the bucket level and verify account- or organization-level controls.
4. Inspect effective ACLs and policies for backup-and-restore98825501 and validate whether unauthenticated access occurred.
5. Review S3 server access logs or CloudTrail data events for access from identities other than helpdesk.luke after 09:58:01 UTC.
6. Classify the eight accessed objects and begin breach-impact assessment, including legal and privacy review where appropriate.
7. Assume secrets_vault_dump.bak and configuration backups may contain credentials; rotate exposed secrets and keys.
8. Enforce MFA, least privilege, short-lived credentials, and conditional access controls for IAM identities.
9. Preserve CloudTrail, S3 access logs, IAM credential reports, and relevant SIEM exports for incident-response evidence.

19. Analytical Limitations

- The dataset proves event occurrence but does not include the content of retrieved S3 objects.
- The exact credential-acquisition method cannot be confirmed from CloudTrail alone.
- The 185.192.70.x address rotation is consistent with shared infrastructure, but ownership was not established.
- Disabling bucket-level Block Public Access controls does not by itself prove anonymous access; account-level controls and effective policies/ACLs were not supplied.
- No host, endpoint, identity-provider, or network telemetry was available to validate activity outside AWS CloudTrail.

20. Conclusion

The supplied CloudTrail evidence supports a coherent and high-confidence compromise chain. The IAM user helpdesk.luke experienced a rapid sequence of failed console logins followed by success, after which the identity enumerated S3, retrieved eight objects, weakened public-access safeguards on a backup bucket, and created a privileged secondary user.

The investigation demonstrates why cloud forensics must correlate identity, API action, target resource, timestamp, source, response, and error fields. The most consequential analytical distinction was separating ListBuckets discovery from GetObject access. This produced the correct first-access timestamp and a defensible incident timeline.

Appendix A — Query Logic Reference

SPL element	Purpose
index="aws_cloudtrail"	Restricts the search to the supplied AWS CloudTrail dataset.
eventSource	Selects the AWS service that generated the event.
eventName	Identifies the API or console action performed.
userIdentity.userName / arn	Identifies the acting cloud identity.
sourceIPAddress	Shows the reported request origin or AWS service hostname.
requestParameters	Reveals the bucket, object, target user, group, or configuration.
responseElements	Contains returned resources or authentication results.
errorCode / errorMessage	Indicates whether AWS rejected the request and why.
stats	Aggregates repeated events to expose patterns.
sort 0 _time	Sorts the full result set chronologically without the default result limit.

Appendix B — References

- CyberDefenders — AWSRaid: <https://cyberdefenders.org/blueteam-ctf-challenges/awsrail/>
- AWS — What is CloudTrail?: <https://docs.aws.amazon.com/awscloudtrail/latest/userguide/>
- AWS — Understanding CloudTrail events: <https://docs.aws.amazon.com/awscloudtrail/latest/userguide/cloudtrail-events.html>
- AWS — Logging Amazon S3 API calls using CloudTrail: <https://docs.aws.amazon.com/AmazonS3/latest/userguide/cloudtrail-logging.html>
- AWS — Blocking public access to S3 storage: <https://docs.aws.amazon.com/AmazonS3/latest/userguide/access-control-block-public-access.html>
- AWS IAM API — CreateUser: https://docs.aws.amazon.com/IAM/latest/APIReference/API_CreateUser.html
- AWS IAM API — AddUserToGroup: https://docs.aws.amazon.com/IAM/latest/APIReference/API_AddUserToGroup.html
- MITRE ATT&CK — T1110.001 Password Guessing: <https://attack.mitre.org/techniques/T1110/001/>
- MITRE ATT&CK — T1078.004 Valid Accounts: Cloud Accounts: <https://attack.mitre.org/techniques/T1078/004/>
- MITRE ATT&CK — T1619 Cloud Storage Object Discovery: <https://attack.mitre.org/techniques/T1619/>
- MITRE ATT&CK — T1530 Data from Cloud Storage: <https://attack.mitre.org/techniques/T1530/>
- MITRE ATT&CK — T1136.003 Create Account: Cloud Account: <https://attack.mitre.org/techniques/T1136/003/>
- MITRE ATT&CK — T1098.003 Additional Cloud Roles: <https://attack.mitre.org/techniques/T1098/003/>